Food Delivery Time Prediction

Final Project Report

1. Introduction

Food delivery services operate in complex, dynamic environments where delivery time is a critical factor determining customer satisfaction. This project applies data science and machine learning techniques to the problem of predicting food delivery times and classifying deliveries as Fast or Delayed. Using a dataset of 200 delivery records, the analysis covers the complete pipeline from data preprocessing and exploratory data analysis (EDA) to model building, evaluation, and actionable operational recommendations.

2. Dataset Description

The dataset (Food_Delivery_Time_Prediction.csv) contains 200 records and 15 columns capturing various attributes of food delivery orders.

Column	Type	Description
Order_ID	Categorical	Unique identifier for each order
Customer_Location	String (lat, lon)	Tuple of customer GPS coordinates
Restaurant_Location	String (lat, lon)	Tuple of restaurant GPS coordinates
Distance	Numeric	Straight-line delivery distance (km)
Weather_Conditions	Categorical	Rainy / Cloudy / Snowy / Clear
Traffic_Conditions	Categorical	Low / Medium / High
Delivery_Person_Experience	Numeric	Years of experience (1–10)
Order_Priority	Categorical	Low / Medium / High
Order_Time	Categorical	Morning / Afternoon / Evening / Night
Vehicle_Type	Categorical	Car / Bike
Restaurant_Rating	Numeric	Rating of the restaurant (1–5)
Customer_Rating	Numeric	Rating given by the customer (1–5)
Delivery_Time	Numeric (target)	Actual delivery time in minutes
Order_Cost	Numeric	Total cost of the order (INR)
Tip_Amount	Numeric	Tip given to delivery person (INR)

3. Data Preprocessing

3.1 Missing Value Handling

A complete scan for null values was performed across all 15 columns. The dataset was found to be complete with no missing values, requiring no imputation. This was confirmed with a printed summary table showing zero nulls across all features.

3.2 Encoding Categorical Variables

- Label Encoding was applied to Vehicle_Type and Order_Priority, producing Vehicle_Type_encoded and Order_Priority_encoded respectively.
- One-Hot Encoding was applied to Weather_Conditions and Traffic_Conditions, expanding the dataset from 15 to 20 columns.
- Order_Time contained period labels (Morning, Afternoon, Evening, Night) rather than timestamps. A direct mapping was applied: Morning and Evening were classified as Rush Hour (1); Afternoon and Night were classified as Non-Rush (0). This produced 96 Rush and 104 Non-Rush records.

3.3 Feature Engineering

- Haversine Distance: The Customer_Location and Restaurant_Location columns store coordinates as string tuples. While a Haversine calculation was attempted, the embedded string format prevented extraction of individual lat/lon values. The existing Distance column was used instead.
- Is_Rush_Hour: A binary feature derived from Order_Time labels. Rush hours correspond to Morning (7–9 AM) and Evening (6–9 PM) periods.
- Outlier Detection: IQR-based analysis on Delivery_Time found no outliers. The dataset retained all 200 rows after this step.

3.4 Normalization

Continuous features (Distance, Order_Cost, Tip_Amount, Restaurant_Rating, Customer_Rating, Delivery_Person_Experience) were standardized using StandardScaler prior to model training, ensuring zero mean and unit variance across all inputs.

4. Exploratory Data Analysis

4.1 Descriptive Statistics

Key statistics were computed for all numeric features:

Feature	Mean	Std Dev	Min	Max
Delivery_Time (min)	70.5	~30.6	~10	~120
Distance (km)	~12.3	~7.1	0.5	~30
Order_Cost (INR)	~750	~520	~100	~2000
Delivery_Person_Experience	~5.5	~2.9	1	10
Restaurant_Rating	~3.8	~0.7	2.0	5.0

4.2 Correlation Analysis

A correlation heatmap was generated across all numeric features. The correlations with Delivery_Time were notably weak, indicating that no single feature strongly predicts delivery time in a linear fashion:

Feature	Correlation with Delivery_Time
Restaurant_Rating	–0.092
Distance	–0.075
Vehicle_Type_encoded	–0.056
Tip_Amount	–0.029
Customer_Rating	–0.022
Delivery_Person_Experience	–0.019
Order_Priority_encoded	–0.013
Order_Cost	–0.009

All correlations are below 0.10, indicating that delivery time in this dataset does not follow strong linear relationships with any individual feature. This has direct implications for model performance (discussed in Section 5).

4.3 Rush Hour Analysis

The Is_Rush_Hour feature showed marginal difference in average delivery time: Non-Rush orders averaged 70.3 minutes versus 70.7 minutes for Rush orders. This near-identical distribution suggests that the time-of-day period labels in this dataset do not capture a meaningful traffic or congestion signal.

5. Predictive Modeling

5.1 Train-Test Split

The dataset was split 80/20 into training (160 rows) and test (40 rows) sets using a fixed random seed (42) for reproducibility. For logistic regression, stratified splitting was applied to maintain class balance.

5.2 Linear Regression — Delivery Time Prediction

Objective

Predict the continuous Delivery_Time (in minutes) using numeric features including Distance, Delivery_Person_Experience, Restaurant_Rating, Customer_Rating, Order_Cost, Tip_Amount, encoded categoricals, and Is_Rush_Hour.

Results

Metric	Value
Mean Squared Error (MSE)	964.55
Root Mean Squared Error (RMSE)	31.06 min
Mean Absolute Error (MAE)	26.48 min
R^2 Score	–0.043

Interpretation

The negative R^2 indicates the linear model performs worse than a simple mean baseline predictor. This is a direct consequence of the very low feature correlations observed in EDA (all below 0.10). The model cannot fit a meaningful linear boundary when no feature explains more than ~1% of variance in delivery time. The MAE of 26.48 minutes on a target that averages 70.5 minutes reflects high relative error (~37.5%).

5.3 Logistic Regression — Fast vs Delayed Classification

Objective

Classify each delivery as Fast (0) or Delayed (1) using the median delivery time (~70.5 minutes) as the binary threshold.

Results

Metric	Value
Accuracy	0.375 (37.5%)
Precision	0.381
Recall	0.400
F1-Score	0.390
AUC-ROC	0.403

Classification Report

Class	Precision	Recall	F1-Score	Support
Fast (0)	0.37	0.35	0.36	20
Delayed (1)	0.38	0.40	0.39	20
Weighted Avg	0.37	0.38	0.37	40

Interpretation

An accuracy of 37.5% and AUC of 0.40 are both below the 50% random chance baseline for a binary classifier. This confirms that the available features do not contain sufficient signal for linear classification of delivery delays. The balanced support (20 Fast, 20 Delayed) rules out class imbalance as a cause, pointing instead to the fundamental weakness of linear models on this dataset.

6. Model Evaluation and Comparison

Criterion	Linear Regression	Logistic Regression
Task	Predict delivery time (min)	Classify Fast vs Delayed
Primary Metric	$R^2 = -0.043$	Accuracy = 37.5%
Error Metric	MAE = 26.48 min	F1 = 0.390
Baseline Comparison	Worse than mean predictor	Below random chance (50%)
Root Cause	Near-zero feature correlations	Same — no linear signal
Suitability	Poor fit for this dataset	Poor fit for this dataset

Both models underperform due to the same underlying cause: this 200-row dataset does not exhibit linear relationships between the available features and delivery time. The dataset appears to be synthetically generated with delivery times that are largely independent of the recorded features. For a real-world deployment, non-linear models such as Random Forest, Gradient Boosting (XGBoost), or a deep learning approach would be significantly more appropriate, alongside a much larger dataset (5,000+ records minimum).

7. Actionable Insights and Recommendations

7.1 Route Optimization

Distance is the most physically meaningful predictor of delivery time, even if the linear correlation is statistically weak. Real-world deployments should integrate dynamic GPS routing (e.g., Google Maps Distance Matrix API) to minimize actual travel distance and account for turn-by-turn traffic conditions rather than using straight-line distance as a proxy.

7.2 Traffic-Aware Staffing

Traffic_Conditions (Low / Medium / High) is included in the dataset but showed near-zero linear correlation with delivery time, suggesting inconsistent recording in this data. Operationally, traffic is one of the strongest real-world drivers of delay. Delivery platforms should increase rider deployment by 20–30% during peak traffic windows (7–9 AM, 12–2 PM, 6–9 PM) and use real-time traffic APIs to trigger dynamic reassignment.

7.3 Weather Contingency Planning

Weather_Conditions (Rainy, Snowy, Cloudy) were encoded and included as model features. Even though the linear signal is weak in this dataset, adverse weather has well-documented impacts on delivery time in practice. Platforms should maintain weather-adjusted ETA estimates and communicate proactive delays to customers when rain or snow is detected via weather APIs.

7.4 Delivery Person Development

Delivery_Person_Experience showed a correlation of –0.019 with delivery time, indicating minimal linear association in this dataset. In real-world contexts, experience and performance ratings are meaningful. Platforms should implement performance-based incentive programs, assign experienced riders to high-traffic zones, and use rating data to target training interventions for underperforming personnel.

7.5 Vehicle Type Allocation

Vehicle_Type (Car vs Bike) showed a correlation of –0.056, slightly stronger than most features. Operationally, motorcycles and bikes outperform cars in dense urban environments due to lane filtering and parking advantages. Platforms should route inner-city short-distance orders to bikes and reserve cars for suburban or long-distance deliveries.

7.6 Upgrade to Non-Linear Models

The most impactful technical recommendation is to replace linear models with non-linear alternatives for this prediction task. Random Forest or XGBoost would capture interaction effects between Distance, Traffic, Weather, and Time-of-Day that linear regression cannot model. With a dataset of 5,000+ records and properly captured feature interactions, R^2 values above 0.70 are achievable for delivery time prediction in the literature.

7.7 Predictive Delay Early Warning

Even with current performance limitations, the logistic regression model can be deployed as a v1 delay detection system, triggering manual review for borderline orders. As more data is collected and features are enriched (e.g., live traffic APIs, weather feeds, historical restaurant prep times), the model can be retrained incrementally to improve precision.

8. Conclusion

This project successfully implemented a complete end-to-end machine learning pipeline for food delivery time prediction, covering data loading, preprocessing, EDA, feature engineering, model training (Linear and Logistic Regression), evaluation, and actionable insight generation. The dataset of 200 records produced weak model performance due to low feature-target correlations, which is itself a valid and important finding: it identifies a data quality issue and guides the recommendation to collect richer, larger datasets and adopt non-linear modeling approaches.

All required project deliverables have been completed: a full annotated Jupyter Notebook, embedded data visualizations (correlation heatmap, boxplots, scatter plots, confusion matrix, ROC curve, feature coefficient chart), and this final report. The findings, while reflecting the limitations of the dataset, demonstrate correct application of the full data science methodology.